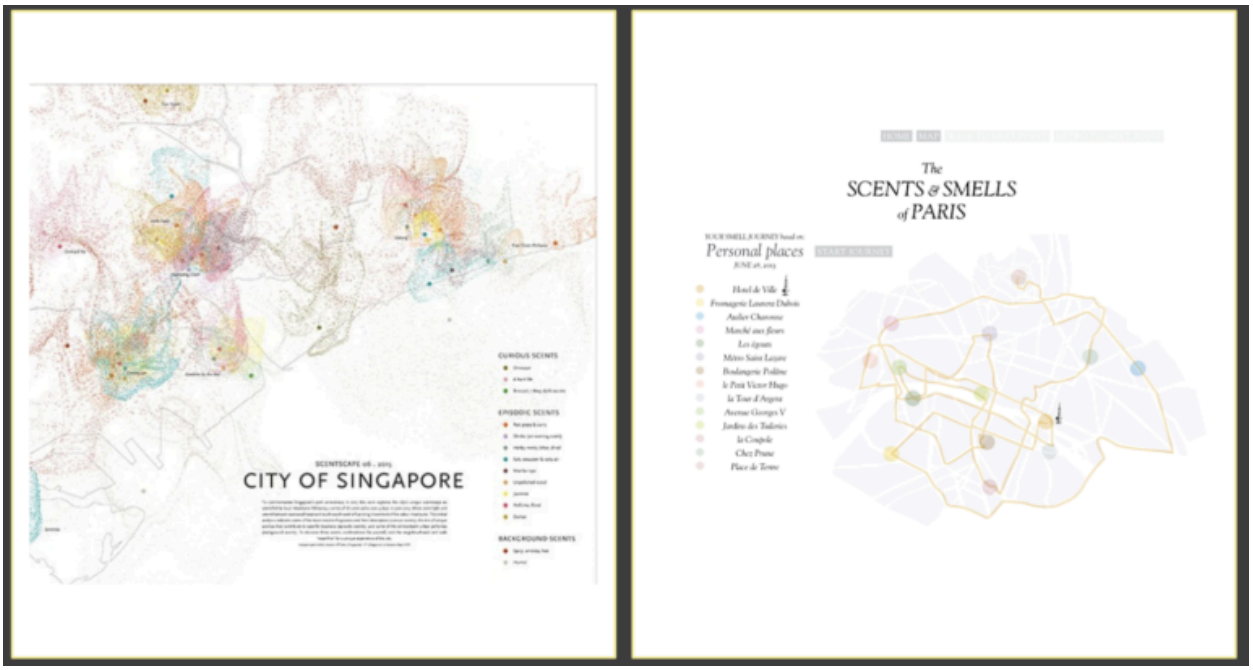
GRENOBLE INP UGA Génie industriel

Rapport

DATA COLLECTION CAMPAIGN: BUILDING A SMELL CARTOGRAPHY



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SMART ANALYTICS FOR BIG DATA

2025 - 2026

Smell Cartography of Grenoble, Saint-Martin-d'Hères & Échirolles

1. Data Collection Strategy:

Source of the data	**Sensors**	**Paid humans**	**Smell-mapping mobile game with rewards**	**Volunteers (citizen scientists, community groups)**	**Smell Inspector IX-16**
Type of data	Air Quality (NO2, PM2,5), VOCs, Humidity, Temperature	Smell Walkers Input	Data from the users	Personal Input	Device for standardized smell logging
Associated V's	***Volume:*** They generate large amounts of numerical data continuously ***Velocity:*** Data streams in real time enabling dynamic monitoring	***Veracity:*** Trained individuals provide high-quality and consistent observations using standardized protocols ***Value:*** Very costly, but strategically valuable to get accurate information	***Volume:*** Depends on the number of users but can generate thousands of reports ***Variety:*** The information submitted can be text, emojis, GPS coordinates or even audio, which will enrich the sensory dataset.	***Veracity:*** Community members can offer deep local knowledge that's very trustworthy	***Velocity:*** 64-channel data captured every 1.8 seconds) ***Veracity:*** Standardized instrument that can minimize human bias ***Value:*** ideal for scientific validation of the other inputs from various resources but will be very costly
Strategies to collect Data	Deploy low-cost IoT sensor networks across diverse neighborhoods or integrate with existing public networks.	Recruit diverse assessors via local job platforms while compensating fairly; Offer training using neutral smell lexicons. And also we should rotate routes to avoid bias.	Develop a gamified app like "Smell Quest" with badges, leaderboards, and local rewards like café or supermarket vouchers	Partner with neighborhood associations or schools and provide simple toolkits and training.	Deploy as a field tool for researchers and volunteers

Location	Specific Areas at a time, until we cover the entire city eventually	Specific Areas that are very hard to access or that are very popular to keep the cost down	Can cover the whole city depending on the users	Cover the whole city if possible, depends on the number of volunteers	Very specific areas to get accurate readings
Real Life Example	PurpleAir Integrated with Atmo Auvergne-Rhône-Alpes network.	**https://www.indeed.com/q-smell-tester-jobs.html**	https://share.google/mqFwSveMy14dXl0vO		www.smart-nanotubes.com

2. Discussion:

To build a reliable and inclusive smell map of Grenoble, Saint-Martin-d'Hères, and Échirolles, I combined several types of data, each chosen for what it does best.

First, sensors like PurpleAir units connected to the Atmo Auvergne-Rhône-Alpes network, collect continuous real-time data on air quality, humidity, and pollutants. This gives us objective, high-volume information that helps explain why certain smells appear; For example, traffic fumes near tram lines or fresh air near the Isère river.

Second, paid smell walkers act like trained observers. Inspired by real quality assurance roles like those at fragrance or deodorant companies, they follow clear guidelines to describe smells in a neutral way. They focus on key or hard-to-reach areas, and their input is highly accurate, making it valuable even if we use them in fewer locations.

Third, a mobile game with rewards (like "Smell Quest") invites everyday people to share what they smell using text, emojis, photos, or GPS coordinates. This method can cover the whole city if enough people play, and it brings in diverse voices especially younger residents or those who might not join a formal study.

Fourth, volunteers from schools, neighborhood groups, or cultural centers add local knowledge that machines can't capture. For example, they might note the smell of roasting chestnuts during winter markets or fresh bread from a family bakery. Their reports are trustworthy because they come from lived experience.

Finally, a tool like the Smell Inspector IX-16, based on real e-nose technologies, can take very fast, standardized readings in specific spots. It doesn't replace human input but helps check whether other data (like citizen reports or sensor spikes) match actual odor conditions.

Together, these sources create a balanced map: sensors show the environment, trained observers ensure quality, games and volunteers bring in community voices, and the inspector device adds scientific validation. The result is a smell cartography that is not only accurate but also fair, respectful, and truly representative of how people experience the city every day.